



Chetan Soni during presentation at IEW 2025, Bengaluru

and estimate the time required for each one. This may include the power supply section, the microcontroller section, or any other functional block. Effort is typically estimated in hours, for example, four or five hours per block.

Based on these estimates, a consolidated plan is created that defines the total project effort, say around seventy hours. Whether eight hours a day are spent working or more is secondary. What matters is having a realistic and structured timeline. Time for reading datasheets is always included and cannot be skipped. Without a proper datasheet study, there is no design, and every necessary activity must be part of the schedule.

Once the plan is defined, it must be followed. If specific tasks are committed for a given day, they must be completed. If the schedule slips, delays compound and critical details get missed. I also keep the project manager or reporting manager regularly updated. During the design phase, there may be dependencies on inputs from other teams. When that happens, it is communicated immediately along with the impact on the timeline. If a day is lost waiting for information, the plan is formally updated. A timeline is not a one-time exercise.

It must be tracked and updated continuously to remain effective.

Component selection. Component selection starts early. Even before creating the architecture, some components may already be shortlisted, and in many cases, the architecture itself is built around those choices. Component selection usually continues until the design is frozen, and some changes as requirements evolve are normal.

I make it a point to explore components extensively. This means regularly visiting semiconductor vendor websites and reviewing their portfolios. There are many vendors in the ecosystem, and they continuously introduce new solutions. Limiting selection to a single product or vendor reduces design flexibility. Exploring available devices helps identify components that meet system requirements more efficiently.

In many cases, a function that appears to require multiple components can be implemented using a single integrated IC, reducing system complexity. Once a component is selected, the datasheet must be read thoroughly. This step cannot be skipped.

A real example highlights why this matters. A team had already shipped a system to an OEM, and

out of four thousand units, one unit failed in the field. When the unit was returned and analysed, it was found that the selected component used an internal clock with an accuracy of plus or minus ten per cent. This clock was used to service the watchdog.

The watchdog window was defined as one thousand milliseconds, with a closed window of five hundred milliseconds and an open window of five hundred milliseconds. The watchdog was serviced at five hundred and ten milliseconds, just after the open window began. Most units worked fine, but one failed in the field. The solution was to service the watchdog at five hundred sixty milliseconds instead, after which the microcontroller stopped resetting.

The root cause was clearly documented in the datasheet. With a clock accuracy of plus or minus ten per cent, a five-hundred-millisecond window has a tolerance of fifty milliseconds on either side. This means the open window could start as early as four hundred fifty milliseconds or as late as five hundred fifty milliseconds. This detail was explicitly mentioned in the watchdog section of the datasheet, but was overlooked by both the hardware and software teams.

Because of this oversight, the team had to explain the issue to the customer, leading to unnecessary delays and blame shifting. The lesson is simple. To be a good hardware engineer, the datasheet must be read thoroughly, along with application notes and reference manuals.

When to simulate and how to analyse

Next comes simulation and analysis. Ideally, I should perform simulations wherever possible. However, I also understand that many simulation tools are expensive. Not everyone can afford software for thermal simulation, power delivery network analysis, or signal integrity